

Keywords: video diffusion • physical consistency • seriality gap • autoregressive generation

S2PD: Serial-to-Parallel Diffusion for Physically and Logically Consistent Video Generation

A Two-Phase Diffusion Algorithm Resolves the Seriality Gap by Running Autoregressive Diffusion at High Noise and Parallel Diffusion at Low Noise

By Jeffrey Hu, Daniel Olmeda Reino & Ayush Tewari (Massachusetts Institute of Technology)

arXiv:2610.06847 • October 2026

Bidirectional video diffusion models violate physical laws and logical rules even when trained on unlimited in-distribution data—a structural obstacle called the *seriality gap*. S2PD resolves it with a single algorithmic change: run the denoiser autoregressively at high noise, then switch to parallel denoising at low noise, combining physical consistency with practical sampling speed.

AT A GLANCE

Problem: Bidirectional diffusion cannot reliably coordinate causally dependent video events, even with unlimited training data.

Why it matters: Physics simulations, game environments, and embodied AI all require causally consistent video generation.

Method: S2PD: autoregressive diffusion for the high-noise phase; parallel diffusion for the low-noise phase.

Result: More reliable rule-following than bidirectional baselines; faster sampling than fully autoregressive methods.

Evidence: Games, physical simulations, and real video benchmarks.

The Big Picture

Video diffusion models have reached remarkable visual quality, but a persistent blind spot—the *seriality gap*—limits their utility for scientific simulation, game generation, and embodied AI. Even when trained to convergence on procedurally generated data with perfect ground-truth labels, bidirectional diffusion models continue to violate physical laws: collision outcomes are wrong, logical sequential rules are broken, and causal chains are severed.

Prior work (arXiv:2607.13031) identified and characterized this gap. S2PD is the first method to close it without

sacrificing the sampling speed advantage of diffusion over fully autoregressive approaches.

Why Existing Approaches Fall Short

More denoising steps do not help. The seriality gap is not a capacity issue. For deterministic video prediction, additional denoising steps do not add effective serial computation—each step still processes all frames in parallel.

Architectural width does not help. Wider networks also fail to close the gap, confirming that the bottleneck is the absence of serial computation, not model capacity.

Fully autoregressive generation is slow. Frame-by-frame autoregressive generation provides the necessary serial compute but multiplies sampling time by the number of frames, making it impractical for high-resolution or long video generation.

Core Mechanism

The key insight is that different noise levels require different amounts of serial computation:

- **High noise** ($\sigma > \sigma^*$): the video is essentially pure noise; the denoiser must decide the *global causal structure*—which events happen, in what order, with what consequences. This requires serial computation to propagate causal dependencies frame by frame.
- **Low noise** ($\sigma \leq \sigma^*$): the global structure is already established; the denoiser only needs to refine visual detail, which can be done in parallel without violating consistency.

S2PD switches between autoregressive and parallel denoising at threshold σ^* .

Technical Formulation

Let $\mathbf{x}_\sigma$ be the noisy video at noise level σ , with frames $(x_{\sigma,1}, \dots, x_{\sigma,T})$. The S2PD denoising trajectory is:

Serial phase ($\sigma > \sigma^*$): denoise frame by frame,

$$\hat{x}_{0,t} = \epsilon_\theta(x_{\sigma,t}, \sigma \mid x_{0,1:t-1}), \quad t = 1, \dots, T$$

where conditioning on previously denoised frames is implemented via masked self-attention or cross-attention.

Parallel phase ($\sigma \leq \sigma^*$): denoise all frames jointly,

$$(\hat{x}_{0,1}, \dots, \hat{x}_{0,T}) = \epsilon_\theta(\mathbf{x}_\sigma, \sigma)$$

starting from the latent at the end of the serial phase.

Total sampling time scales as approximately:

$$\tau_{\text{S2PD}} \approx (T - 1) \tau_{\text{serial-step}} + \tau_{\text{parallel}}$$

substantially less than $T \tau_{\text{serial-step}}$ for fully autoregressive generation.

Learning or Inference Procedure

S2PD is a **training-free** inference algorithm applied to any pretrained video diffusion model with autoregressive conditioning support.

1. Sample Gaussian noise $\mathbf{x}_{\sigma_{\max}}$.
2. Run the DDIM solver serially (frame by frame) from $\sigma_{\max}$ to σ^* .
3. Run the solver in parallel over all frames from σ^* to 0.
4. Return the denoised video $\hat{\mathbf{x}}_0$.

The switching threshold σ^* is the only hyperparameter and is set to the midpoint of the noise schedule in all experiments.

What the Guarantee Says

No formal consistency guarantee is provided. The empirical claim is: S2PD produces fewer physical and logical violations than matched bidirectional baselines across all tested environments, while requiring less sampling time than fully autoregressive methods. The intuition is that the high-noise serial phase establishes the causal skeleton, making the low-noise parallel phase sufficient for detail refinement.

Experimental Findings

- **Games:** S2PD follows Tetris collision rules and Sokoban push logic more reliably than bidirectional baselines.
- **Physical simulations:** Lower rule-violation rates on billiards and rigid-body interaction benchmarks.
- **Real video:** Improved temporal stability on real-world video prediction while maintaining visual quality.
- **Sampling efficiency:** Substantially faster than fully autoregressive generation while closing the consistency gap.

Ablations and Interpretation

Switching threshold: Earlier switching (less serial computation) degrades consistency; later switching improves consistency but reduces efficiency gains. The midpoint is Pareto-optimal.

Phase ablations: Removing the parallel phase (fully serial) matches S2PD’s consistency but increases sampling time by $T\times$; removing the serial phase (fully parallel) restores the seriality gap.

Error analysis: Violations concentrate in frames requiring coordination with distant prior frames, confirming that the gap is a long-range causal dependency problem

rather than a local temporal one.

Reference

Jeffrey Hu, Daniel Olmeda Reino & Ayush Tewari. **S2PD: Serial-to-Parallel Diffusion for Physically and Logically Consistent Video Generation**. arXiv:2610.06847, October 2026.
<https://arxiv.org/abs/2610.06847>